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Nutrient and Soil Health Monitoring for Mustard Greens

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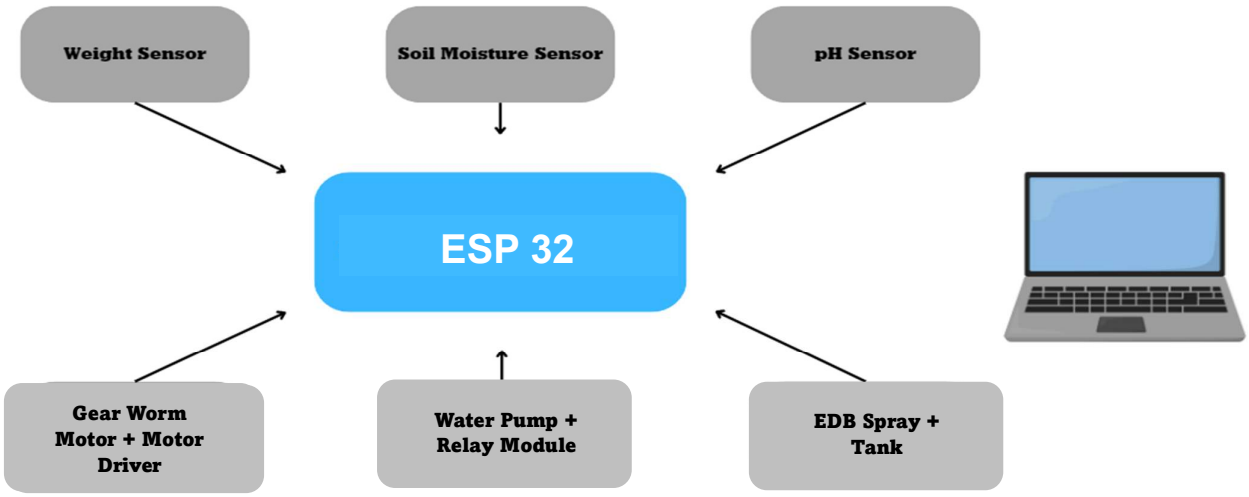
Full Description:

For our project, 'Nutrient and Soil Health Monitoring for Mustard Greens,' the automated fertilizer system will release the appropriate amount of fertilizer. The quantity will be estimated using the weight sensor (HX711) to ensure precise nutrient application and prevent over-fertilization or wastage. The soil moisture sensor module will monitor the soil's moisture level to determine if irrigation is necessary together with the fertilizer, ensuring optimal absorption. The water pump will then deliver the liquid from the tank through the hose to maintain proper hydration.

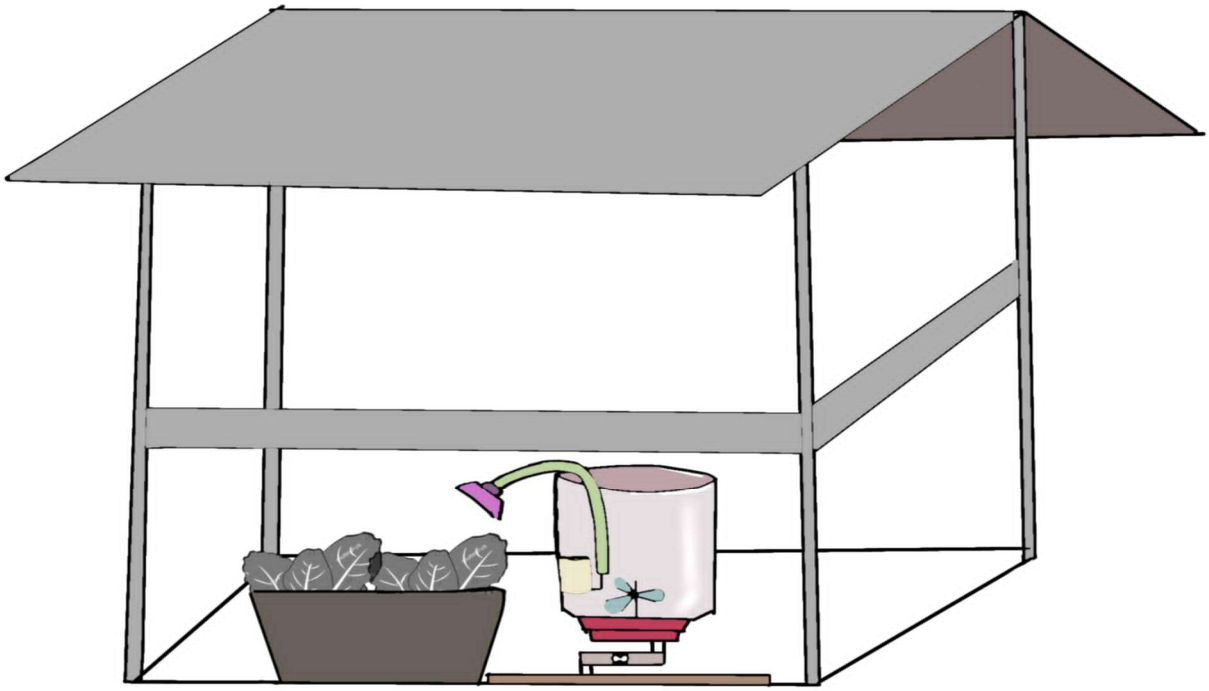
The entire system will be controlled by the ESP32 WROOM, which will process data from the sensors and automate the irrigation and fertilization process. The breadboard and circuit kit will facilitate proper wiring and connections, ensuring the smooth operation of all components. With this system, we can efficiently monitor nutrient levels, soil health, and water distribution, ensuring optimal conditions for Mustard greens plant growth while minimizing waste and manual labor.

Components	Use:
ESP32	The microcontroller that serves as the brain of the system, processing sensor data and controlling various components.
Weight Sensor (HX711) + Load Cell	Measures the amount of fertilizer used, ensuring precise nutrient application and preventing wastage.
Soil Moisture Sensor (Capacitive)	Detects soil moisture levels to determine when irrigation is needed, preventing overwatering or underwatering.
pH Sensor	Monitors soil pH to maintain optimal conditions for tomato growth, ensuring proper nutrient availability.
Gear Worm Motor + Motor Driver	Powers mechanical component to mix the fertilizer.
8-hole Nozzle	A spraying mechanism that applies water and nutrient solutions evenly to the soil or leaves.
Hose	Channels water from the tank to the soil or plant roots.
Tank	Stores liquid for irrigation and distribution.
Water Pump + Relay Module	Pumps water from the storage tank to the irrigation system, ensuring a steady and controlled water supply for the plants.
Bowl	Used to mix or collect fertilizers.

Block Diagram of Nutrient and Soil Health Monitoring for Mustard Greens



Prototype Concept Sketch



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